

The Effect of Small-World Network Topology on Algorithmic Polarization Thresholds in a Bounded Confidence Model

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1 Abstract

Algorithmic content filtering reduces cross-opinion interaction, thereby accelerating opinion fragmentation and polarization; yet the threshold for polarization being structurally inevitable remains unknown. 19 papers were reviewed across 5 areas: foundational bounded confidence models, algorithmic bias extension, structured network topologies, empirical filter bubble evidence, and context within Australian policy.

In fully connected populations, foundational models establish the consensus threshold $\epsilon^* = 0.5$. Algorithmic bias extension papers show that algorithmic filtering does exacerbate fragmentation, but assume mean field connectivity. Studies on scale free and random networks show that topology affects polarization outcomes, but without clustering as an independent variable. The Watts-Strogatz model fills this gap, as its rewiring parameter controls clustering.

The effect of clustering on the algorithmic polarization is conceptually grounded but untested within continuous opinion dynamics.

2 Keywords

Opinion dynamics, bounded confidence models, algorithmic bias, small-world networks, echo chambers, polarization threshold, social network topology

3 Introduction

Social media algorithms selectively expose users to content users already agree with, reducing cross opinion discourse that opinion dynamics models need to reach consensus. As regulators, including the Australian Competition and Consumer Commission, develop obligations for digital platforms, quantitative evidence on algorithmic harm thresholds is absent from the discussion.

The proposed research addresses this gap by investigating whether the clustering structure of social networks changes the algorithmic bias threshold γ^* at which opinion polarization becomes unavoidable. The question lies within complex systems simulation, with secondary relevance to network science and computational social science. Agent based modeling on structured graphs is a well established approach for studying collective behavior. Bounded confidence models have extensive theoretical exploration; what is unanswered is the interaction between algorithmic filtering strength and network clustering coefficient.

This literature review will address this main question: what is known about the effect of network topology and clustering on polarization thresholds in algorithmic BCM models, and where does the literature leave a gap?

4 Materials and Methods

4.1 Search strategy

Literature was identified through 3 search streams in March-April 2026. Google Scholar searched with keyword combinations drawn from model vocabulary: “bounded confidence model”, “algorithmic bias”, “opinion dynamics”, “Watts-Strogatz”, “small-world network” and “polarization threshold”. Semantic Scholar was used to go along citation graphs from two key papers; Sirbu et al. (2019) and Pansanella et al. (2022a) to identify foundational predecessors and recent extensions. ArXiv and SSRN were searched for pre-prints in cases where peer-reviewed publication of recent (and credible) relevant work was not yet confirmed. No date range restriction was applied, and searches were restricted to English language sources.

4.2 Inclusion and exclusion criteria

Papers were included if they addressed at least one of 3 core criteria: (1) Bounded confidence model or extension, (2) effect of network topology on opinion dynamics, (3) algorithmic content filtering and effect on polarization. Peer reviewed papers were preferred, but pre-prints were included when authors already had publication records in network science or computational social science. 3 pre-prints were excluded due to insufficient author credibility or different model families. Papers using only discrete or binary opinions were included only if their topological findings were applicable to continuous BCM setting.

4.3 Resulting corpus

The screening process yielded 19 papers across 5 categories: foundational BCM theory (4 papers), algorithmic bias in mean-field models (4 papers), algorithmic bias on structured networks (5 papers), BCM on Watts-Strogatz topology without algorithmic bias (3 papers) and empirical or policy context (3 papers).

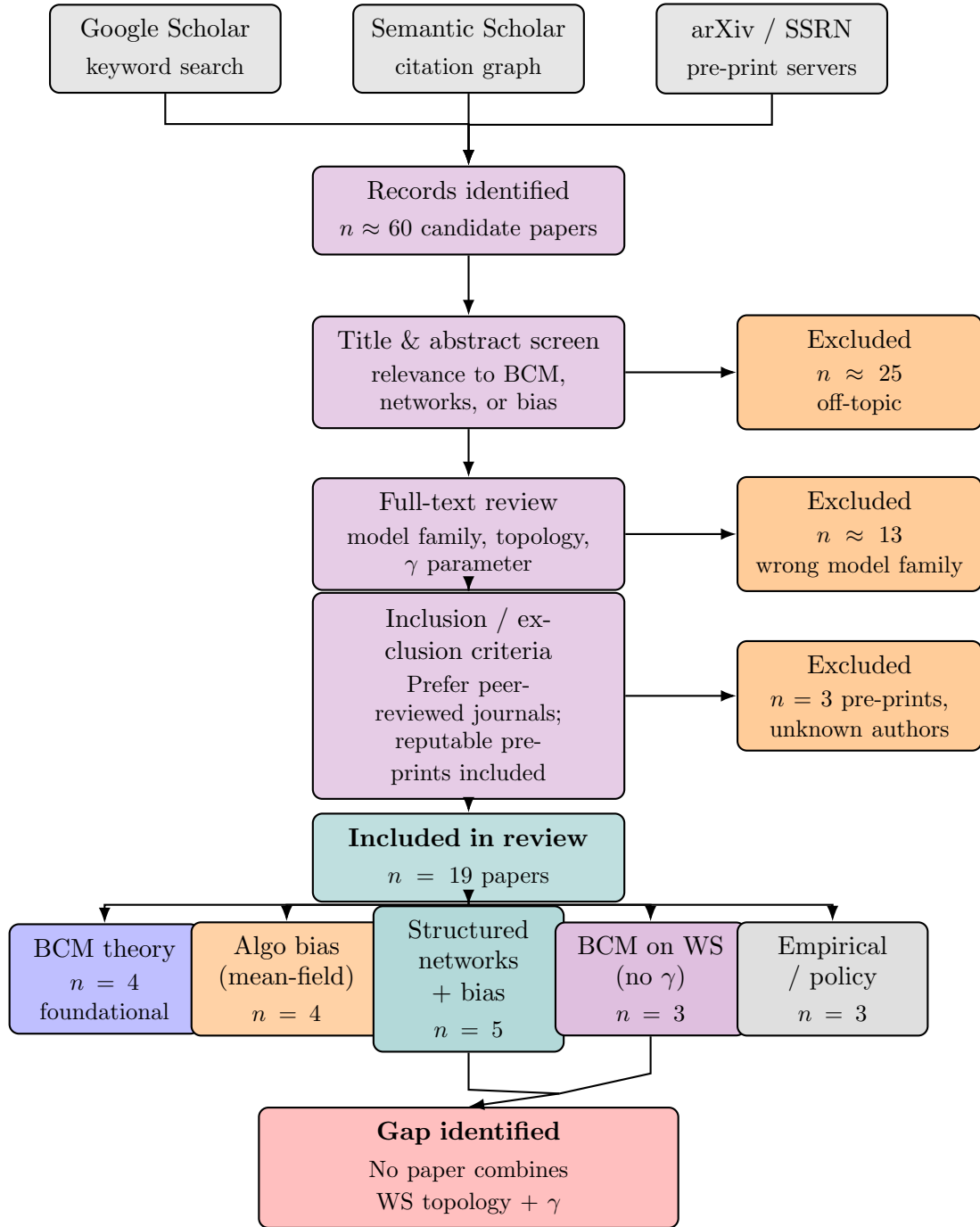


Figure 1: Literature review screening and categorization process (PRISMA-like).

4.4 Validity and replicability

Repeating these queries on Semantic Scholar from the same anchor papers should result in the same corpus. To extend this review in the future, the recommended procedure is to run forward citation searches from Pansanella et al. (2022a) and Sîrbu et al. (2019), as these are the papers most likely to accumulate relevant citations. Thematic categorization was cross-checked using each paper’s self described contribution.

5 Results

Literature shows a line from mean-field BCM to structured network extensions, with algorithmic bias introduced as a tunable parameter at an intermediate stage. 19 papers consistently find that both algorithmic filtering and network topology shape polarization outcomes, but no study tests their interaction with clustering as an **independent** variable.

5.1 Foundational bounded confidence models

The Deffuant-Weisbuch (DW) model established the framework for continuous opinion dynamics (Deffuant et al., 2000). Agents hold opinions on a unit interval and update two at a time when their opinion distance falls within a *confidence bound* ϵ . Analytical papers show that there is a sharp consensus threshold at $\epsilon^* = 0.5$; below this value, opinions fragment into stable clusters, and above it the population converges to a single opinion. The Hegselmann-Krause (HK) model introduced a synchronous update variable, in which each agent averages over all neighbors within its confidence bound, thereby recovering similar threshold behavior (Hegselmann and Krause, 2002). However, both models assume that every agent can interact with each other, which limits their applicability to real social networks.

Castellano et al. (2009) introduced a statistical physics framing that situates the threshold within the broader scope of phase transitions; the consensus/fragmentation boundary is akin to a thermodynamic critical point. Thus the appropriate question is not whether polarization increases gradually, but whether a sharp discontinuity exists as a control parameter crosses a critical value. This framing is the direct motivation for the research design, which seeks out γ^* as a phase boundary in (γ, ϵ) space. The 2024 survey by

Bernardo et al. (2024) consolidates 20 years of convergence theory, confirming that thresholds are consistent across DW and HK variants and providing the technical vocabulary used throughout this review.

5.2 Algorithmic bias as tunable parameter

Sîrbu et al. (2019) introduced the key theoretical extension of an algorithmic bias parameter γ to the mean-field DW model. The probability under this formulation that agent i interacts with agent j is proportional to the inverse opinion distance to the power of γ . Higher values of γ increasingly concentrate interactions among like-minded agents, modeling algorithm behavior. The key finding is that raising γ raises the fragmentation level, even at confidence bounds above the $\gamma = 0$ consensus threshold, meaning that algorithmic filtering can produce polarization in conditions where an otherwise unfiltered population would reach consensus.

Peralta et al. (2021b) extended this framework, extrapolating equations that result in a complex phase diagram with consensus, coexistence and polarization phases as functions of bias strength and initial opinion distribution. An asymmetric bias variant, where the platform favors one opinion, was found to be capable of determining the final global opinion state entirely (Peralta et al., 2021a). Brooks et al. (2024) independently introduced a sigmoidal BCM with a scalar bias parameter that interpolates between hard threshold BCM and full averaging, and derived conditions for the stability of polarized states on specific graph topologies. Together, these results confirm that the (γ, ϵ) phase diagram is a well-defined object and that γ^* is a meaningful estimable quantity.

Empirical literature provides support as well. Chitra and Musco (2019) demonstrated on real Reddit and Twitter networks that even small algorithm-driven changes to interaction probabilities produce measurable echo chamber formation, and that platforms that optimize for reduced disagreement increase polarization. This connects the simulation models and real-world relevance.

5.3 Effects of network topology on polarization

Pansanella et al. (2022a) represent the closest existing work to the proposed study. Their paper implements the Sîrbu BCM onto the Barabási-Albert and Erdős-Rényi random networks. They found that topology meaningfully

modifies both the severity and speed of fragmentation under algorithmic bias. Scale-free networks, characterized by a small number of high-degree hub nodes, produce earlier and more severe fragmentation than random networks at the same γ value. A companion paper introduced adaptive rewiring, finding that the co-formation of opinions and network structure did not substantially alter final opinion states when rewiring is slow relative to opinion dynamics, supporting the use of static network topologies in simulation designs (Pansanella et al., 2022b).

5.4 Clustering as an independent variable

Foundational characterization of small-world networks by Watts and Strogatz (1998) demonstrated that real networks share two structural properties: high local clustering (most of a node’s neighbors are connected to each other), and a short average path length (any two nodes are reachable in a small number of steps). The Watts-Strogatz model achieves this by starting from a regular ring lattice with high clustering, then progressively rewiring edges with probability p , smoothly transitioning between a lattice and ER random graph. Most importantly, degree distribution remains approximately the same throughout, meaning every node retains approximately k neighbors; p controls *clustering* while holding degree constant.

These properties uniquely suit the Watts-Strogatz (WS) topology to the proposed simulation. Neither scale-free nor random networks allow clustering to be varied in isolation; WS provides the experimental control that existing literature lacks.

Gandica et al. (2010) demonstrated that the DW bounded confidence model is directly applicable to WS networks and produces topology-sensitive outcomes: the number of final opinion clusters varies with rewiring parameter p even in the absence of algorithmic bias. Schawe et al. (2021) extended this using the HK variant, identifying bridge nodes as the structural mechanism that maintains consensus in networked populations. They confirmed that the consensus threshold shifts systematically across ER, BA, WS, and lattice topologies through finite-size scaling analysis.

Perra and Rocha (2019) directly tested algorithmic personalization across multiple topologies including Watts-Strogatz, finding that the clustering coefficient (not hub structure) is the primary driver of echo chamber formation. Although their model uses binary rather than continuous opinion, the topological finding transfers directly to BCM setting as a theoretical prediction.

More clustered networks should polarize at a lower algorithmic bias, implying that γ^* is a decreasing function of clustering coefficient.

5.5 Empirical and policy context

The Australian policy context provides real-world motivation. The Australian Competition and Consumer Commission (2023) Digital Platform Services Inquiry has identified algorithmic recommendations harm as a priority regulatory concern, with government support for new platform obligations confirmed in principle. Nasim et al. (2022) ran a longitudinal analysis of Australian Twitter communities, providing empirical confirmation that echo chamber formation is observable in domestic online communities across contentious topics, and that algorithmic recommendations moderate its severity. These findings establish that polarization is not theoretical, but is shaping public discourse in the Australian context.

5.6 Summary of findings

Table 1 summarizes the 19 included papers across the 5 thematic categories, documenting for each paper whether it uses the DW or HK model family, whether algorithmic bias is implemented, which network topology is modeled, and whether clustering coefficient is isolated as an independent variable.

Paper	Model	γ	Topology	Clustering isolated
Deffuant et al. (2000)	DW	No	Mean-field	—
Hegselmann and Krause (2002)	HK	No	Mean-field	—
Castellano et al. (2009)	Review	No	Various	—
Bernardo et al. (2024)	DW/HK	No	Various	—
Sîrbu et al. (2019)	DW	Yes	Mean-field	—
Peralta et al. (2021a)	DW	Yes	Community	No
Peralta et al. (2021b)	DW	Yes	Community	No
Brooks et al. (2024)	Sigmoidal	Yes	Limited	No
Pansanella et al. (2022a)	DW	Yes	BA, ER	No
Pansanella et al. (2022b)	DW	Yes	BA, ER	No
Watts and Strogatz (1998)	—	—	WS	Yes
Barabási and Albert (1999)	—	—	BA	No
Gandica et al. (2010)	DW	No	WS	Partial
Schawe et al. (2021)	HK	No	WS, ER, BA	Partial
Serracín Morales and Nettasinghe (2026)	Affective	No	WS, BA	Partial
Perra and Rocha (2019)	Binary	Yes	WS, BA, ER	Partial
Chitra and Musco (2019)	FJ	Yes	Real-world	No
Nasim et al. (2022)	Observational	—	Twitter	—
Australian Competition and Consumer Commission (2023)	Policy	—	—	—

Table 1: Summary of included papers. *Clustering isolated* denotes whether clustering coefficient was varied independently of degree distribution. Partial indicates WS topology was used but γ was absent.

6 Discussion

The literature reviewed agrees on a clear picture that algorithmic bias and network topology both independently drive opinion polarization in BCM, but their interaction has not been tested with clustering as an isolated variable. The proposed study addresses this, and the theoretical motivation for expecting a cluster-dependent polarization threshold is well-supported across multiple model families and empirical contexts.

6.1 Principal results

There are 3 findings from the reviewed papers that are of direct consequence to the proposed research.

The consensus threshold in the unbiased DW model is established at $\epsilon^* = 0.5$ (Deffuant et al., 2000), providing a replicable validation target. The existence of this phase transition motivates the search for a similar critical value γ^* under algorithmic bias, and confirms that the (γ, ϵ) phase diagram is a well-defined object.

The Sîrbu et al. (2019) extension shows that algorithmic bias shifts the fragmentation boundary in mean-field populations, producing polarization at otherwise consensus-reaching confidence bounds. The analytical extensions by Peralta et al. (2021a,b) and Brooks et al. (2024) confirm that this boundary is topology dependent. The implication is that γ^* is not a universal constant, but an unknown function of network structure.

The papers on BCM dynamics involving structured networks consistently show that Watts-Strogatz topology produces different outcomes from both scale-free and random networks, even in the absence of algorithmic bias. Gandica et al. (2010) demonstrated this for the DW model, Schawe et al. (2021) provided a mechanistic explanation through bridge nodes, and Perra and Rocha (2019) found clustering coefficient to be the primary structural driver of echo chamber formation. The convergence of these findings across independent studies and model families constitutes strong theoretical motivation for this study.

6.2 Limitations

Several limitations of the reviewed literature have consequences on the interpretation of the proposed study’s results.

The closest existing work (Pansanella et al., 2022a) does not report confidence intervals or effect sizes for topology-dependent differences in the polarization threshold. This makes it difficult to predict the severity of the effect the proposed study is trying to detect. If the topology effect is small relative to the variance across simulation runs, 50 runs per condition may be insufficient to estimate γ^* reliably.

Additionally, most analytical results in the algorithmic bias literature assume symmetric initial opinion distributions, or specific structures that may not generalize to the uniform and bimodal distributions used in the proposed parameter sweep. The phase diagrams derived by Peralta et al. (2021b) are more indicative, rather than predictive, for the DW model on WS topology.

While the empirical literature confirms that echo chambers form in real

networks (Chitra and Musco, 2019; Nasim et al., 2022), this does not provide ground truth values for γ that would allow simulation results to be calibrated against observed platform behavior. The proposed study can identify thresholds in a formal model, but cannot directly translate those thresholds into specific algorithmic design parameters.

6.3 Proposed future work

The immediate extension of the proposed study is to test adaptive WS networks, where edges rewire based on opinion similarity with algorithmic bias. Pansanella et al. (2022b) found that adaptive rewiring has limited effect on final states in BA and ER networks, but this result has not been tested in WS topology.

A second extension would be to vary the degree parameter k in the WS construction, separating the effect of clustering from the effect of mean degree. The proposed study holds k at a constant of 6, meaning that clustering and local connectivity are not disconnected. Varying k across a range of p values could allow a more complete picture of the γ^* driving factors.

6.4 Comparison with prior work

This study bridges 2 existing lines of research. Sirbu et al. (2019) and Pansanella et al. (2022a) establish that algorithmic bias drives polarization and that network topology modifies this effect. Gandica et al. (2010) and Schawe et al. (2021) establish that BCM dynamics on WS networks are topology sensitive, without algorithmic bias. By combining both lines of study by embedding the Sirbu model on WS networks, this work will contribute an estimate of how the clustering coefficient shifts the polarization threshold in a continuous opinion dynamics setting.

In this respect, this study closely resembles Pansanella et al. (2022a) in methodology agent-based simulation with parameter sweeps, benchmarked against mean-field results, but extending the topology range in a theoretically motivated direction.

6.5 Conclusion

The reviewed literature establishes that the effect of network clustering on the polarization threshold is well motivated, and empirically suggested by

adjacent model families. No existing study provides this test within the continuous BCM framework. If the clustering hypothesis holds: that higher clustering lowers γ^* , the result would provide quantitative evidence that social network structure determines algorithmic harm, with implications for how platform obligations are designed and enforced (Australian Competition and Consumer Commission, 2023).

7 Self-Reflection

This literature review utilized AI search tools like Semantic Scholar alongside traditional search engines like Google Scholar. While it was a great help in identifying very specific papers that would not be able to be accounted for by simple pattern matching, there was always possibility of missing out on papers that should have been included based on the criteria. Generative AI writing tools helped lay the foundation and outline, and gave me structure to write the paragraphs in a self-contained manner.

References

- Australian Competition and Consumer Commission. Digital platform services inquiry 2020–25. <https://www.accc.gov.au/inquiries-and-consultations/finalised-inquiries/digital-platform-services-inquiry-2020-25>, 2023. Accessed: 20 May 2026.*.
- A.-L. Barabási and R. Albert. Emergence of scaling in random networks. *Science*, 286(5439):509–512, 1999. <https://doi.org/10.1126/science.286.5439.509>. *.
- C. Bernardo, C. Altafini, A. Proskurnikov, and F. Vasca. Bounded confidence opinion dynamics: A survey. *Automatica*, 159:111302, 2024. <https://doi.org/10.1016/j.automatica.2023.111302>. *.
- H. Z. Brooks, P. S. Chodrow, and M. A. Porter. Emergence of polarization in a sigmoidal bounded-confidence model of opinion dynamics. *SIAM Journal on Applied Dynamical Systems*, 2024. <https://doi.org/10.1137/22M1527258>. *.

- C. Castellano, S. Fortunato, and V. Loreto. Statistical physics of social dynamics. *Reviews of Modern Physics*, 81(2):591–646, 2009. <https://doi.org/10.1103/RevModPhys.81.591>. *
- U. Chitra and C. Musco. Understanding filter bubbles and polarization in social networks. arXiv preprint, 2019. *
- G. Deffuant, D. Neau, F. Amblard, and G. Weisbuch. Mixing beliefs among interacting agents. *Advances in Complex Systems*, 3(1–4):87–98, 2000. <https://doi.org/10.1142/S0219525900000078>. *
- Y. Gandica, M. del Castillo-Mussot, G. J. Vázquez, and S. Rojas. Continuous opinion model in small-world directed networks. *Physica A: Statistical Mechanics and its Applications*, 389(24):5864–5870, 2010. <https://doi.org/10.1016/j.physa.2010.08.025>. *
- R. Hegselmann and U. Krause. Opinion dynamics and bounded confidence: Models, analysis and simulation. *Journal of Artificial Societies and Social Simulation*, 5(3), 2002. URL <https://jasss.soc.surrey.ac.uk/5/3/2/2.pdf>. *
- M. Nasim, D. Weber, T. South, J. Tuke, N. Bean, L. Falzon, and L. Mitchell. Are we always in strife? A longitudinal study of the echo chamber effect in the Australian Twittersphere. arXiv preprint, 2022. *
- V. Pansanella, G. Rossetti, and L. Milli. From mean-field to complex topologies: Network effects on the algorithmic bias model. In R. M. Benito, C. Cherifi, H. Cherifi, E. Moro, L. M. Rocha, and M. Sales-Pardo, editors, *Complex Networks & Their Applications X*, Studies in Computational Intelligence, pages 329–340, Cham, 2022a. Springer International Publishing. https://doi.org/10.1007/978-3-030-93413-2_28. *
- V. Pansanella, G. Rossetti, and L. Milli. Modeling algorithmic bias: simplicial complexes and evolving network topologies. *Applied Network Science*, 7(1):57, 2022b. <https://doi.org/10.1007/s41109-022-00495-7>. *
- A. F. Peralta, J. Kertész, and G. Iñiguez. Opinion formation on social networks with algorithmic bias: dynamics and bias imbalance. *Journal of Physics: Complexity*, 2(4):045009, 2021a. <https://doi.org/10.1088/2632-072X/ac340f>. *

- A. F. Peralta, M. Neri, J. Kertész, and G. Iñiguez. Effect of algorithmic bias and network structure on coexistence, consensus, and polarization of opinions. *Physical Review E*, 104(4):044312, 2021b. <https://doi.org/10.1103/PhysRevE.104.044312>. *.
- N. Perra and L. E. C. Rocha. Modelling opinion dynamics in the age of algorithmic personalisation. *Scientific Reports*, 9(1):7261, 2019. <https://doi.org/10.1038/s41598-019-43830-2>. *.
- H. Schawe, S. Fontaine, and L. Hernández. When network bridges foster consensus. Bounded confidence models in networked societies. *Physical Review Research*, 3(2):023208, 2021. <https://doi.org/10.1103/PhysRevResearch.3.023208>. *.
- A. Serracín Morales and B. Nettasinghe. Affective polarization on small-world and scale-free networks. arXiv preprint arXiv:2603.27845, 2026. *.
- A. Sîrbu, D. Pedreschi, F. Giannotti, and J. Kertész. Algorithmic bias amplifies opinion fragmentation and polarization: A bounded confidence model. *PLOS ONE*, 14(3):e0213246, 2019. <https://doi.org/10.1371/journal.pone.0213246>. *.
- D. J. Watts and S. H. Strogatz. Collective dynamics of ‘small-world’ networks. *Nature*, 393(6684):440–442, 1998. <https://doi.org/10.1038/30918>. *.